

Toufiq Musah

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Education

University of Pennsylvania
M.S.E Data Science

Aug 2025 – Present

Relevant Courses: Machine Learning for Data Science, Mathematical Foundations for Machine Learning, Computer Systems Programming, Big Data Analytics

Kwame Nkrumah University of Science and Technology (KNUST)
BSc. Biomedical Engineering – First Class Honors

Jan 2021 – Nov 2024

Thesis Topic: Explainable Artificial Intelligence Framework for Clinical Decision Support Systems in Stroke Triaging
Relevant Courses: Research Methods, Biomechanics, Medical Imaging, Biosignal Processing and Analysis, Probability and Statistics, Linear Algebra, Multivariable Calculus, C/C++, Biomaterials, Bioinstrumentation,

Relevant Experience

Kumasi Centre for Collaborative Research in Tropical Medicine (KCCR) Kumasi, Ghana

Oct 2024 – Oct 2025

- Research assistant and engineer in the Global Health and Infectious Diseases Group
- Applied research in deep learning, computer vision, and medical image analysis
- Development of context-specific Large Language Model tools for disease management

Responsible Artificial Intelligence Lab (RAIL)

Kumasi, Ghana

Oct 2023 – Feb 2024

- Research assistant in the Responsible Artificial Intelligence Lab
- Applied research in machine learning for biomedical engineering and medical imaging applications
- Generative modeling for CT denoising, super-resolution, and cross-modality CT–MRI translation

Research and Academic Work

Publications

Toufiq Musah, Salvatore Calcagno, Federica Proietto Salanitri, Xiaomeng Li, Maruf Adewole, Marawan Elbatel ‘**ULF-Synth: Physics-Guided Ultra-Low-Field MRI Enhancement for Pediatric Neuroimaging**’, MICCAI AFRICAI Workshop, 2026.

Toufiq Musah, Prince Ebenezer Adjei, ‘**Multimodal Computed Tomography Radiomics for Stroke Outcome Prediction**’, Medical Image Understanding and Analysis Conference, 2026.

Toufiq Musah, ‘**Large Kernel MedNeXt for Breast Tumor Segmentation and Self-normalizing Network for pCR Classification in Magnetic Resonance Images**’, MICCAI Deep-Breath Workshop, MAMA-MIA Challenge, 2025.

Toufiq Musah, Chinasa Kalaiwo, et. al. ‘**Towards Trustworthy Breast Tumor Segmentation in Ultrasound using Monte Carlo Dropout and Deep Ensembles for Epistemic Uncertainty Estimation**’, MICCAI MIRASOL Workshop, 2025.

Claudia Takyi Ankomah, Livingstone Eli Ayivor, ... and **Toufiq Musah**. ‘**How We Won BraTS-SSA 2025: Brain Tumor Segmentation in the Sub-Saharan African Population Using Segmentation-Aware Data Augmentation and Model Ensembling**’, MICCAI BraTS Challenge, 2025.

Toufiq Musah, Philip Nkwam, Ajay Sharma. ‘**Automated Pediatric Brain Hippocampal and Basal Ganglia Segmentation in Ultra-low Field Magnetic Resonance Images**’, MICCAI LISA Challenge, 2025.

Toufiq Musah, Prince Ebenezer. Adjei, Kojo Obed Otoo, ‘**Automated Segmentation of Ischemic Stroke Lesions in Non-Contrast Computed Tomography Images for Enhanced Treatment and Prognosis**’, MICCAI Meets Africa Workshop, 2024 🏆.

Selected Talks & Orals

[Surgical Video Scene Understanding & Panoptic Segmentation](#), African Computer Vision Summer School, 2026

[Automated Medical Image Segmentation: Toward Trustworthy Clinical Tools](#), COMPASS, 2025

[Towards Trustworthy Breast Tumor Segmentation](#), MIRASOL Workshop, MICCAI Conference, 2025

[Large Kernel MedNeXt for Breast Tumor Segmentation](#), MIDL Young Researcher Showcase, 2025

[Optimisation Techniques in Machine Learning](#), Ghana Data Science Summit, Tutorial, 2025

Relevant Projects

Multi-Modal Data Integration for Head and Neck Tumor Prognostication

- Developed a multimodal learning framework to predict 5-year survivability and recurrence in head and neck tumor cases using whole-slide image features, structured clinical data, laboratory analytes, surgical notes.
- Generated tiny ClinicalBERT embeddings from free-text notes and fused modalities via a low-rank multimodal bilinear fusion, achieving an AUC of 0.75 across prognostic tasks.

Vision-Language Modeling for Surgical Scene Understanding & Panoptic Segmentation

- Surgical scene understanding using surgical video and panoptic segmentations as multimodal inputs.
- Trained S²-Scaled SwinUNETR for panoptic segmentation across 21 classes, enabling precise spatial localization.
- Fine-tuned MedGemma with QLoRA for scene graph generation and automated detection of key surgical events.

Collaborative AI Resources for Expectant Mothers (C.A.R.E)

- Developed an accessible LLM-powered assistant using Retrieval-Augmented Generation to support expectant mothers managing gestational diabetes.
- Deployed the system through WhatsApp to maximize accessibility in resource-limited settings.

Generative AI for Medical Imaging Enhancement and Data Augmentation

- Designed a super-resolution and denoising pipeline to generate high-quality head CT scans from low-resolution images to reduce effective radiation exposure.
- Implemented a Deep Convolutional Generative Adversarial Network for synthetic medical image generation for unsupervised pre-training and efficient downstream finetuning.

Awards and Acknowledgments

- Best Poster Presentation Award - Deep Learning Indaba, 2025
- First Place - Brain Tumor Segmentation Challenge (SSA), 2025
- Best Poster Presentation Award - Ghana Data Science Summit, IndabaX, 2024
- Best Oral Presentation Award - MICCAI Meets Africa Workshop, 2024
- MasterCard Healthcare Entrepreneurship Grant (\$10,000)
- MICCAI Education Challenge Finalist, 2024
- MICCAI Travel Award, 2024, 2025

Volunteering and Community Activities

Team Coordinator, Sprint AI Training for Medical Image Knowledge Translation 2024 – Present

Developing training materials and coordinating the [Sprint AI training program](#). Building capacity of emerging researchers and clinicians in medical image analysis and deep learning applications.

Challenge Organizer – Technical Committee, Breast Ultrasound Analysis Challenge 2025

Led the technical organization of the [PRECISE-Abreast Challenge](#), Breast Ultrasound Segmentation and Classification. Built end-to-end submission and automated evaluation workflows for rigorous and reproducible model assessments.

Course Facilitator, Women in Engineering – SheCodes Club 2023 – 2026

Teaching Python programming and introductory machine learning classes for the Women in Engineering Society, creating an inclusive learning environment to help members gain technical skills and build their confidence in STEM.

IndabaX, Ghana Data Science Summit 2024, 2025

Contributing to the organization of [Ghana Data Science Summit](#). Worked with various committees to promote the event. Hosted an engaging tutorial session at the event.

Blog Posts, Tutorials

Surgical Video Scene Understanding and Scene Graph Generation with MedGemma – [Medium Article](#)

A Hitchhiker's Guide to 3D Medical Image Processing – [Medium Article](#)

Introduction to TorchIO for 3D MRI Processing – [Preprocessing \(Part 1\)](#) - [Augmentation & Dataloaders \(Part 2\)](#)

End-to-End Deep Learning Tutorial for Image Classification: Pneumonia Detection – [Colab Notebook](#)

Skills

Languages: Python, LaTeX, C/C++, MATLAB | **Libraries:** PyTorch, TensorFlow, Scikit-Learn